500ºF THERMALLY CONDUCTIVE ADHESIVE

For High Power Applications

500ºF DURALCO™ 132

Just Mix and Apply

Cures at Room Temperature

Provides Fast Heat Transfer

Ideal for any Industrial, Electrical or Electronic Application

Duralco thermally conductive adhesives combine Cotronics' unique, high temperature resins with highly conductive fillers to form thermally conductive, adhesive bonds with continuous service up to 500ºF.

Easy to use. Just mix and apply.

100% solids formulations. No volatiles. No VOC's.

Duralco 132 has excellent adhesion to metals, glass, ceramics, and plastics.

Offers excellent resistance to chemicals solvents and moisture.

Provides the heat dissipation required for many high temperature electronic and industrial applications.

Users Report:

- Duralco™ 132 dissipates heat in a semiconductor device.
- Duralco™ 132 transfers the heat generated in high power devices and provides for efficient cooling.
- Duralco™ 132 bonds electrical heating elements for fluid heating.

Applications include: bonding and assembling heating coils, cooling coils, heating elements, heat sinks, reaction vessels, semi-conductors, rectifiers, power supplies, replacement for soldering and welding, etc.

Duralco™ 132P is commonly used as a heat tracing adhesive which is used to bond heating or cooling tubing to equipment. Its' non sag formulation and room temperature curing, provides for the most efficient placement of the heating or cooling coils.

Duralco™ 132 is the ideal choice for high temperature, high power electronic or industrial applications requiring high thermal conductivity.

Availability:

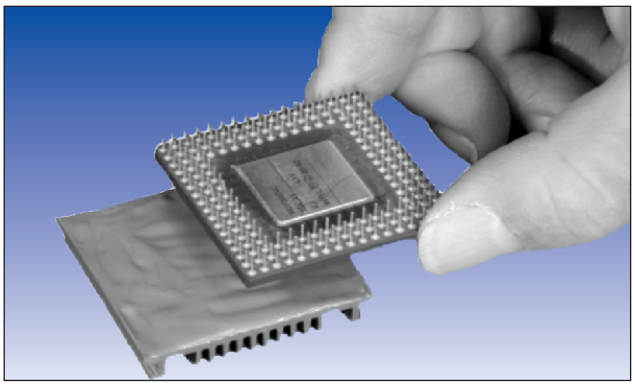
Cat. No.	Description	Temp.
Duralco™ 132-1	16 oz. Kit	500°F
Duralco™ 132-2	32 oz. Kit	500°F
Duralco™ 132IP-1	16 oz. Kit	500°F
Duralco™ 132IP-2	32 oz. Kit	500°F
Duralco™ 132P	16 oz. Kit	500°F
Duralco™ 132P-IP	16 oz. Putty	500°F

Pre-Measured Kits

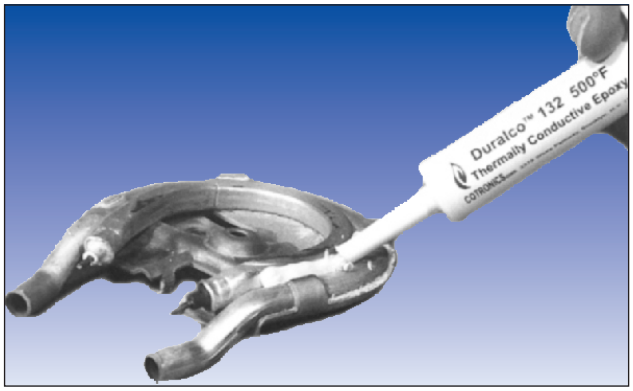
Each Unit Contains: *1 jar of resin, 1 syringe of hardener and 1 1 mixing stick. (See page 19 for details)*

EE 132-10	10 - 10gm units/box
EE 132-25	10 - 25gm units/box
EE 132IP-10	10 - 10gm units/box
EE 132IP-25	10 - 25gm units/box

Quantity Prices & Custom Formulations Available Upon Request



Duralco™ 132 Dissipates Heat in A Semi Conductor Device



Duralco™ 132 Bonds Electrical Heating Element for Fluid Heating

Physical Properties Maximum Temperature	132 500°F	132IP 500°F
Components - Color 2	Silver	Silver
Mixed Viscosity (cps)	14,500	43,000
Hardness (Shore D)	75	75
Flexural Strength (psi)	1,150	1,600
Compressive Strength (psi)	6,000	6,350
Thermal Conductivity (BTU-in/Hr. Ft^2-°F)	30	30
Thermal Expansion (10^{-5}/°C)	4.1	4.1
Volume Resistivity (ohm-cm)	10^6	10^6
Heat Distortion (°C)	210	210
Thermal Stability (%1000 hr @ 200°C)	0.2	0.2
Shrinkage (% max.)	0.8	0.8
Moisture Absorption (% 30 Days)	0.2	0.2
Mix Ratio (by wt.)	100:8	100:8
Cure (Hrs. @ R.T.)	16-24*	16-24*

* Cures can be accelerated with mild heat.

www.cotronics.com | Phone: 718-788-5533 | Facsimile: 718-788-5538

600ºF THERMALLY CONDUCTIVE ADHESIVES

For Electrical and Industrial Applications

Duralco Thermally Conductive adhesives and potting compounds provide the heat dissipation required for many High Temperature Electronic and Industrial applications.

These ultra temp. adhesives combine Cotronics' unique polymer system and special thermally conductive fillers to provide continuous service up to 650ºF.

Duralco Conductive Adhesives have excellent adhesion to glass, ceramics, metals and plastics.

Resistant to most chemicals and solvents.

They are ideal for all high temperature applications.

Applications Include:

- Removing the heat generated in many electronic applications including semi-conductors, rectifiers, high power devices, etc.
- Heat transfer applications, including bonding copper coils to reaction vessels for heating and or cooling. Heat tracing adhesive. etc.
- Fabrication of heated, plastic forming tools, molds, etc.

Duralco ™ Maximum Temperature	128 500°F	132 500°F	133 600°F	134 500°F	135 500°F
Major Constituent	**Ceramic**	**Aluminum**	**Aluminum**	**Ceramic**	**Aluminum**
Features	Hi Electrical Resistance	Hi Thermal Conductivity	High Temperature	Electrically Resistant Grease	Thermally Conductive Grease
Volume Resistance (ohm-cm.)	10^{15}	10^{6}	10^{6}	10^{16}	N. A.
Thermal Conductivity (BTU-in/Hr. Ft^2-°F)	20	30	40	35	40
Mixed Viscosity (cps)	79,000	15,000	36,500	Grease	Grease
Color	Tan	Silver	Silver	Tan	Gray
No. of Components	2	2	2	1	1
Mix Ratio	100/5	100/8	100/30	N. A.	N. A.
Cure Cycle (Hrs. @ 75°F)	16 - 24*	16 - 24*	4 hrs. @ 250	N. A.	N. A.

* Cures can be accelerated with mild heat.

Duralco 132 Dissipates Heat in a Semiconductor Device

Duralco™ 128 - 500ºF Ceramic Based

Duralco™128 is highly thermally conductive, electrically resistant adhesive and potting compound.

The ceramic fillers are carefully chosen to provide high thermal conductivity and high dielectric strength.

Just mix the resin and hardener, apply and cure at room temp. Curing may be accelerated with mild heat.

Duralco™ 132 - 500ºF Aluminum Based

Duralco™ 132 is an Aluminum Metal Filled Epoxy that cures at room temperature to form machinable, thermally conductive bond lines.

Duralco™ 132 provides the maximum heat transfer available in a 500°F epoxy system.

Can be supplied as a non-sag putty, Duralco™ 132P, for heat tracing applications.

Users report 132 bonds copper coils to vessels for rapid heating and cooling in a critical chemical process.

Used for bonding, assembling and heat tacking applications.

Cat. No.	Description	Temp.
Duralco™ 128	8 oz.	500°F
Duralco™ 132-1	16 oz.	500°F
Duralco™ 132-2	32 oz.	500°F
Duralco™ 132IP-1	16 oz.	500°F
Duralco™ 132P	16 oz.	500°F
Duralco™ 132P-IP	16 oz.	500°F
Duralco™ 133	16 oz.	600°F
Duralco™ 134	8 oz.	500°F
Duralco™ 135	4 oz.	500°F
Duralco™ 135	8 oz.	500°F

Quantity Prices, Special Packing Upon Request

Duralco™ 133 - 600ºF Aluminum Based

Duralco™ 133 is a two component, heat curing, Aluminum Filled, Conductive Epoxy.

Duralco™ 133 combines the excellent properties of Duralco™ 132 with Cotronics' higher temp. epoxy systems making it suitable for applications requiring up to 600°F service.

Cures with mild heat to form thermally conductive bond lines and heat transfer medium.

Duralco™ 133 is suitable for high temperature tooling. It is readily machinable and ideal for all kinds of repairs and as a construction material.

Duralco™ 134 - 500ºF Ceramic Based Grease

Duralco™ 134 thermally conductive grease, is a non-hardening, electrically insulating and thermally conductive grease. It is ideal for use between components and heat sinks.

Users Report Duralco™ 134 replaced silicone based grease in manufacturing of high end digital cameras.

Duralco™ 134 retains its paste like consistency, enabling parts to be easily removed and replaced. Will not dry out even after long periods of time. Usable to 500°F.

Duralco™ 135 - 500ºF Aluminum Based Grease

Duralco™ 135 is filled with an ultra fine, aluminum metal powder to provide the maximum possible heat transfer rate in a non hardening grease.

Used in delicate military applications where excess heat build up can cause serious failures.

Duralco™ 135 is commonly used in many industrial applications where electrical resistance is not critical.